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Semiconductor placing in packaging

Abstract

A method for placing a semiconductor onto a substrate is provided. The method includes the following steps: transferring, using a placement tool, the semiconductor along a path over onto the substrate; lowering, using the placement tool, the semiconductor to a predetermined height above the substrate; titling, using the placement tool, the semiconductor, to a predetermined angle; determining, using the placement tool, a first contact point of the semiconductor to the substrate at the predetermined angle; determining, using the placement tool, the first contact point is shift-off from an alignment position on the semiconductor with respect to the substrate; adjusting, using the placement tool, the first contact point to correct the shift-off; and lowering, using the placement tool, the semiconductor to make a first contact with the substrate at the corrected first contact point.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7146718	12/2005	Thuerlemann	414/744.8	H01L 21/67144
11255529	12/2021	Wu	N/A	H01L 24/02
2014/0311652	12/2013	Kostner	156/60	H01L 24/75
2023/0392921	12/2022	Yoshida	N/A	H01L 24/75

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Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS (1) The present application claims priority to U.S. Provisional Patent Application No. 63/274,924, filed on Nov. 2, 2021, the entire disclosure of which is incorporated herein by reference.

TECHNICAL FIELD

(1) The present disclosure relates to hybrid bonding, and more particularly to metal layer structures for reducing dishing and erosion effects.

BACKGROUND

(2) The manufacturing of integrated circuits often involves the bonding of device dies to package substrates. In a typical bonding process, a device die is first picked up from a wafer that has already been sawed into dies. The device die is placed on a table. A pick and place tool then picks up the device die from the table, and then places the device die on a package substrate. After a plurality of devices dies are placed on a plurality of package substrate, the package substrate strip along with the device dies go through a reflow process, so that the device dies are bonded to the package substrates.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure is best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the

industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIGS. 1A-1B are cross-sectional views illustrating various stages of forming an example semiconductor device of interest to the present disclosure.

(3) FIGS. 2A-2B illustrate a placement tool for packing the semiconductor device shown in FIG. 1 according to an embodiment.

(4) FIG. 3 illustrates one example of two layers of semiconductors are placed on a wafer in accordance with the present disclosure.

(5) FIG. 4 illustrates an observed misalignment of semiconductor and semiconductor wafer by a placement tool.

(6) FIGS. 5A-5D illustrate one example of placing a semiconductor using a placement tool in accordance with the disclosure.

(7) FIGS. 6A-6B are simplified cross-sectional views of indentations during placement of a semiconductor using a placement tool in accordance with the disclosure.

(8) FIGS. 7A-7B illustrate embodiments for detecting contact locations using the placement tool in accordance with the present disclosure.

(9) FIG. 8 illustrates an example method for forming a semiconductor package having a bonded structure in accordance with the disclosure.

DETAILED DESCRIPTION

(10) The following disclosure provides many different embodiments or examples for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(11) Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. Prepositions, such as “on” and “side” (as in “sidewall”) are defined with respect, to the conventional plane or surface being on the top surface of the wafer or substrate, regardless of the orientation of the wafer or substrate. The term “horizontal” is defined as a plane parallel to the conventional plane or surface of a wafer or substrate, regardless of the orientation of the wafer or substrate. The term “vertical” refers to a direction perpendicular to the horizontal as defined above, i.e., perpendicular to the surface of a substrate. The terms “first,” “second,” “third,” and “fourth” may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms are only used to distinguish one element, component, region, layer or section from another region, layer or section. Thus, a first element, component, region, layer or section discussed below could be termed a second element, component, region, layer or section without departing from the teachings of the present disclosure.

(12) There are many packaging technologies to house the semiconductors such as the 2D fan-out (chip-first) IC integration, 2D flip chip IC integration, PoP (package-on-package), SiP (system-in-

package) or heterogeneous integration, 2D fan-out (chip-last) IC integration, 2.1D flip chip IC integration, 2.1D flip chip IC integration with bridges, 2.1D fan-out IC integration with bridges, 2.3D fan-out (chip-first) IC integration, 2.3D flip chip IC integration, 2.3D fan-out (chip-last) IC integration, 2.5D (solder bump) IC integration, 2.5D (μ bump) IC integration, μ bump 3D IC integration, μ bump chiplets 3D IC integration, bumpless 3D IC integration, bumpless chiplets 3D IC integration, SoIC and/or any other packaging technologies. It should be understood various embodiments disclosed herein although are described and illustrated in a context of a specific semiconductor packaging technology, it is not intended to limit the present disclosure only to that packaging technology. One skilled in the art would understand those embodiments may be applied in other semiconductor technologies in accordance with principles, concepts, motivations, and/or insights provided by the present disclosure.

(13) System on integrated chip (SoIC) is a recent development in advanced packaging technologies. SoIC technology integrates both homogeneous and heterogeneous chiplets into a single System-on-Chip (SoC)—like chip with a smaller footprint and thinner profile, which can be holistically integrated into advanced WLSI (aka CoWoS® service and InFO). From external appearance, the newly integrated chip is just like a general SoC chip yet embedded with desired and heterogeneously integrated functionalities. SoIC realizes 3D chiplets integration with additional advantages in performance, power and form factor. Among many other features, the SoIC features ultra-high-density-vertical stacking for high performance, low power, and min RLC (resistance-inductance-capacitance). SoIC integrates active and passive chips into a new integrated-SoC system to achieve better form factor and performance. US Patent Publication #20200168527, entitled “SoIC chip architecture” provides some descriptions about some example SoIC structures. US Patent Publication #20200168527 is incorporated by reference in its entirety. Another example of SoIC can be found at <https://3dfabric.tsmc.com/english/dedicatedFoundry/technology/SoIC.htm>, which is also incorporated by reference in the present disclosure in its entirety.

(14) Numerous benefits and advantages are achieved by way of the present disclosure over conventional techniques. For example, embodiments provide an improved placement tool for semiconductor packaging such as chip on wafer (CoW), wafer on wafer (WoW), and/or any other bonded structure. In various embodiments, the placement tool in accordance with the present disclosure includes a head configured to be tilt-able. For placing an individual die on a wafer, the head is tilted to form an angle with respect to an upper surface of the wafer. Before placing the die onto the wafer, the placement tool in accordance with the present disclosure is configured to tilt the head and detect a contact point of the die with the upper surface of the wafer. The placement tool in accordance with the present disclosure is configured to determine whether the contact point is align with a position on the wafer where the die is supposed to be placed. If the placement tool determines a misalignment exists, it is configured to adjust a position of the head to align the die to the position on the wafer where the die is supposed to be placed. Once determining that the die is aligned, the placement tool in accordance with the present disclosure is configured to lay down the die onto the wafer at the contact first and determine whether the die is laid at the contact. Once determining the die has made the contact with the wafer, the placement tool in accordance with the present disclosure is configured to lay down the rest of the die onto the wafer. In this way, precision of placing individual dies onto wafer is improved. These and other embodiments of the disclosure, along with many of its advantages and features, are described in more detail in conjunction with the text below and corresponding figures.

(15) Example Semiconductor Device

(16) FIGS. 1A and 1B are cross-sectional views illustrating various stages of forming an example semiconductor device of interest to the present disclosure. FIG. 1A shows a cross-sectional view of a portion of a first semiconductor **10** and a portion of a semiconductor wafer **20** according to an embodiment. The first semiconductor **10** includes a substrate **101**, and the second semiconductor wafer **20** includes a substrate **201**. In an embodiment, each of the substrates **101** and **201** may

include silicon or other semiconductor materials. In another embodiment, each the substrates **101** and **201** may include other elementary semiconductor materials, such as germanium. In some embodiments, each the substrates **101** and **201** may include a compound semiconductor, such as silicon carbide, gallium arsenic, indium arsenide, or indium phosphide. In some other embodiments, each the substrates **101** and **201** may include an alloy semiconductor, such as silicon germanium, silicon germanium carbide, gallium arsenic phosphide, or gallium indium phosphide. In some embodiments, the substrate **101** and/or substrate **201** may include an epitaxial layer, e.g., the substrate **101** and/or substrate **201** includes an epitaxial layer overlying a bulk semiconductor. (17) As shown, the first semiconductor **10** includes a device region **102** formed on the substrate **101**. The device region **102** includes a gate structure **103** embedded in a dielectric layer **104**, source/drain regions **105**, and isolation (e.g., shallow trench isolation) structures **106**. The gate structure **103** includes a gate dielectric layer **107**, a gate electrode **108**, and possibly insulating materials **109**. The device region **102** shown in FIG. 1A is merely for illustration only and not limiting. Other structures may be formed in the device region **102**. Other transistors (e.g., FinFETs, NMOS, PMOS transistors) and devices (capacitors, resistors, diodes, inductors, and the like) may also be formed on the substrate **101**.

(18) Referring still to FIG. 1A, the dielectric layer **104** is disposed on the substrate **101** and covering the device region **102**. The first semiconductor **10** also includes a plurality of through-substrate vias (TSVs) **130** in the dielectric layer **104** and extending into the substrate **101**. The TSVs **130** are configured to provide electrical connection to the second semiconductor wafer **20**. It is noted that two TSVs are shown for illustration only, the number of TSVs can be any integer number according to actual applications.

(19) In an embodiment, each TSV can include a liner **131**, a diffusion barrier layer **132**, and a conductive material **133**. The liner **131** may include an insulating material, e.g., oxides or nitrides and may be formed by a plasma enhanced chemical vapor deposition (PECVD) process or other deposition processes. The liner **131** may be a single layer or multi-layers. The diffusion barrier layer **132** may include Ta, TaN, Ti, TiN, CoW, or a combination thereof. In an embodiment, the diffusion barrier layer **132** is formed by a physical vapor deposition (PVD) process. The conductive material **133** may include copper (Cu), copper alloy, aluminum (Al), aluminum alloys, or combinations thereof. Alternatively, other applicable materials may also be used. In an embodiment, the conductive material **133** is formed by plating.

(20) The first semiconductor **10** further includes a metallization structure **140** on the TSV **130** and the device region **102** to connect the TSV **130** to the device region **102**. In an embodiment, the metallization structure **140** includes an interconnect structure, such as contact plugs **141** and conductive features **142**. The conductive features **142** are embedded in an insulating material **109**. In some embodiment, the insulating material **109** includes multiple layers of a dielectric material, such as an oxide, e.g., silicon oxide, the contact plugs **141** include copper, aluminum, tungsten, combinations thereof, or the like, and the conductive features **142** include a metallic material, such as copper, copper alloy, aluminum, aluminum alloy, or combinations thereof.

(21) The first semiconductor **10** further includes a bonding structure **150** on the metallization structure **140**. In some embodiments, the bonding structure **150** includes a barrier layer **151** and a conductive material **152**. The barrier layer **151** and the conductive material **152** are embedded in a bonding layer **110** disposed on the insulating material **109**. In some embodiments, the bonding layer **110** includes an oxide or polymer material. The conductive material **152** includes a metallic material, such as copper, copper alloy, aluminum, aluminum alloy, or combinations thereof. When the conductive material **152** includes copper, which can diffuse into the insulating material **109**, the barrier layer **151** is formed between the conductive material **152** and the insulating material **109**. The barrier layer **151** may include silicon nitride (SiN), silicon oxynitride (SiON), titanium nitride (TiN), tantalum nitride (TaN), aluminum nitride (AlN), TaN, Ta/TaN, CoP, CoW, or the like. In some embodiments, the bonding layer **110** includes a polymer material, such as benzocyclobutene

(BCB) polymer, polyimide (PI), or polybenzoxazole (PBO). In some embodiments, the polymer material is deposited over the substrate by spin coating.

(22) The second semiconductor wafer **20** includes a device region **202** on the substrate **201**. The device region is formed in the second semiconductor wafer **20** in a front-end-of-line (FEOL) process. In some embodiments, the device region includes a gate structure **203** embedded in a dielectric layer **204**, source/drain regions **205**, and isolation structures **206**. The gate structure **203** includes a gate dielectric layer **207**, a gate electrode **208**, and spacers **209**. It is noted that the gate structure **203** is merely an example, and other structures may be formed in the gate structure **203**. In some embodiment, the gate structure **203** may include various N-type metal oxide semiconductor (NMOS) and/or P-type metal oxide semiconductor (PMOS) devices, fin-type field-effect transistors (FinFETs), gate-all-around (GAA) devices, memories, and the like. Other devices, such as capacitors, diodes, resistors, photo-diodes, and the like can also be formed on the substrate **201**.

(23) The second semiconductor wafer **20** further includes a metallization structure **240** and a bonding structure **250**. The metallization structure **240** includes contact plugs **241** embedded in a dielectric layer **222** and conductive features **242** embedded in an insulating material **209**. The bonding structure **250** is similar to the bonding structure **150** and includes a barrier layer **251** and a conductive material **252** embedded in a polymer material **210**, such as benzocyclobutene (BCB) polymer, polyimide (PI), or polybenzoxazole (PBO). The barrier layer **251** is similar to the barrier layer **151** and may include silicon nitride (SiN), silicon oxynitride (SiON), titanium nitride (TiN), tantalum nitride (TaN), aluminum nitride (AlN), TaN, Ta/TaN, CoP, CoW, or the like. The conductive material **252** is similar to the conductive material **152** and includes a metallic material, such as copper, copper alloy, aluminum, aluminum alloy, or combinations thereof. A polishing, e.g., a chemical mechanical polishing (CMP), process is performed on the surface of the bonding layers **110**, **210**, of the first and second semiconductor wafers **10** and **20**, respectively.

(24) FIG. 1B shows a cross-sectional view of the first semiconductor **10** and a portion of the second semiconductor wafer **20** of FIG. 1A after an alignment between the two and a bonding of the two are performed according to an embodiment. In an embodiment, the first semiconductor **10** and the second semiconductor wafer **20** are hybrid bonded together by applying pressure and heat to form a stacked structure **30**. In an exemplary embodiment, the hybrid bonding is performed at a temperature in a range between about 100° C. and 200° C., so that the polymer materials **110** and **210** become a non-confined viscous liquid and are reflowed. Thereafter, the stacked structure **30** is further heated to a higher temperature in a range between about 200° C. and about 400° C., so that the conductive materials **152** and **252** are interconnected by thermal compression bonding and polymer materials **110** and **220** are fully cured. In some embodiments, the pressure for hybrid bonding is in a range between about 0.7 bar to about 10 bar. The hybrid bonding process may be performed in an inert environment, e.g., with an inert gas including N.sub.2, Ar, He, or combinations thereof.

(25) Hybrid bonding involves at least two types of bonding, such as metal-to-metal bonding and non-metal-to-non-metal bonding. During a CMP process, corrosion of a copper or copper alloy layer or copper dishing may occur, i.e., a portion of the conductive material **152** and portion of the conductive material **252** may be removed causing a decrease in the electrical interconnection between the first and second conductor wafers **10** and **20**.

(26) Die Placement

(27) Attention is now directed to FIGS. 2A-B, where a placement tool **200** is shown to place individual semiconductor **10** onto a semiconductor wafer **20** shown in FIGS. 1A-B is illustrated. It will be described with reference to FIGS. 1A-B. As shown in FIG. 2A, the placement tool **200** is configured to transfer individual semiconductor **10** from a substrate strip **202** to semiconductor wafer **20**. In various implement, the placement tool **200** includes a head, a stepper motor, a controller, and/or any other components. In those implementation, the controller typically operates

under the control of an operating system and executes or otherwise relies upon various computer software applications, components, programs, objects, modules, engines, data structures. In general, the controller may be configured to control the operation of the head of the placement tool **200** when its instructions are executed by the processor, in order to pick up individual semiconductors **10** and transfer individual semiconductor **10** from the substrate **202**.

(28) As shown in FIG. 2B, in accordance with the present disclosure, the placement tool **200** includes a robotic arm **204** and a pick-up head **206**. The pick-up head **206** is equipped with an acquisition device, such as pneumatic suction cups, capable of temporarily and releasably holding the semiconductor **10**. The robotic arm **204** is configured to move the pick-up head **206** over a motion path **210** shown in FIG. 2A. The motion path **210** may originate at the substrate strip **202** where an individual semiconductor **10** is picked by the pick-up head **206** and terminate at a location on the semiconductor wafer **20**. The motion of the pick-up head **206** may be unbroken and continuous over the motion path **210**.

(29) The robotic arm **204** may be a programmable mechanical arm with links connected by joints allowing rotational motion and/or translational displacement of the pick-up head **206**. The robotic arm **204** may be, for example, a three-axis R-Theta robot arm or a selectively compliant articulated robot arm (SCARA). The robotic arm **204** is configured to manipulate and accurately position the pick-up head **206** and to move the pick-up head **206**. In some implementation, as shown here, the movement of the robotic arm is controlled by a stepper motor **208** of the placement tool **200** shown.

(30) In various implementation, the semiconductors **10** are formed by processing a wafer with front-end-of-line processes. The individual semiconductor **10** may be separated from the wafer by mechanical sawing, by scribing and breaking, by laser cutting, or by a different technique. It should be understood that multiple layers of semiconductors **10** may be stacked on the semiconductor wafer **20** in accordance with the present disclosure. That is, the present disclosure is not limited to only one layer of semiconductors **10** being placed on the semiconductor wafer **20**. FIG. 3 illustrates one example of two layers of semiconductors are placed on a wafer in accordance with the present disclosure.

(31) Referring to FIG. 3, in this example, a multi-die structure **300** is formed, which includes a first die **301c**, and a second die **301b**, stacked on top of a portion of wafer **301a**. Each of the first, second dies **301c** and **301b** may include a substrate, an active region including a plurality of active devices (not shown), an interconnect structure **303** formed on the substrate and configured to electrically connect the active region of each die with each other. The interconnect structure **303** may include a plurality of dielectric layers **303a**, metal lines **303b** formed in the dielectric layers **303a**, and vias **303c** connecting metal lines **303b** in different layers. In some embodiments, the dielectric layers **303a** include silicon oxide, silicon nitride, silicon carbide, silicon oxynitride, and/or combinations thereof. In some embodiments, the dielectric layers **303a** may include one or more low-k dielectric layers having low k values. In some embodiments, the k values of the low-k dielectric materials may be lower than about 3.0.

(32) In some embodiments, the dies **301c** and **301b**, and the wafer **301a** are electrically coupled to each other by through substrate vias (TSVs) and through oxide vias (TOVs) **308**. In some embodiments, the die group **30** also includes a bonding layer **317** including an oxide material, e.g., silicon oxide. In some embodiments, the bonding layer **317** may include a plurality of bonding films and electrical connectors **309** having a plurality of solder regions. In some embodiments, the electrical connectors **309** include copper posts, solder caps, and/or electrically conductive bumps **310** configured to electrically coupled to other electronic circuits on a printed circuit board or other substrates. In an embodiment, the stacked dies of the multi-die structure **30** include logic devices, input/output (**10**) devices, processing units, e.g., data processing units, graphics processing unit, application specific integrated circuits (ASIC), field programmable gate arrays (FPGA), other applicable types of devices. In some embodiment, the multi-die structure **300** is a system-on-

integrated circuits (SoIC) device that includes multiple functions. It is understood that the number of dies in the multi-die structure **300** is illustrative only and is chosen for describing the example embodiment and should not be limiting. For example, the multi-die structure **300** can include a single die, two dies, or more than three dies. In some embodiments, the multi-die structure **300** may be bonded to a package substrate (e.g., an interposer, a printed circuit board) through flip-chip bonding using the electrical connectors **309**.

(33) In some embodiments, the dies and wafer **301a-c** are bonded to each other by a hybrid bonding process. In an embodiment, the wafer **301a** has a first bonding surface formed on its upper surface including a first bonding dielectric layer **315a** and a first conductive contact structure **316a**. The second die **301b** has a second bonding surface formed on a bottom of its substrate, the second bonding surface includes a second bonding dielectric layer **315b** and a conductive contact structure **316b**. In an embodiment, the first and second conductive contact structures **316a**, **316b** may be electrically coupled to the interconnect structure **303**. In another embodiment, the first and second conductive contact structures **316a**, **316b** may not be electrically coupled to the interconnect structure **303**. In an embodiment, the wafer **301a** and the second die **301b** are directly hybrid bonded together, such that the first and second conductive contact structures **316a**, **316b** are bonded together, and the first and second bonding dielectric layers **315a**, **315b** are bonded together. In an embodiment, the first and second bonding dielectric layers **315a**, **315b** each include silicon oxide, and the first and second conductive contact structures **316a**, **316b** each include copper.

(34) In an embodiment, the dies also include a seal ring **320** configured to stop cracks generated by stress during the bonding processes and/or the singulation. The seal ring **320** is also configured to prevent water, moisture, and other pollutant from entering the dies. In an embodiment, the seal ring **320** includes copper configured to suppress electromagnetic noise. In an embodiment, the first die **301a** may include a bonding dielectric layer **330** configured to be bonded to a carrier substrate by fusion bonding.

(35) Alignment

(36) As can be seen, for forming a multi-die structure **300** or a bonded structure **30** shown in FIGS. **1** and **3**, individual semiconductors should be placed onto wafer at locations where the conductive regions are aligned, and the dielectric/insulation regions are aligned. However, it is observed that prior art placement tool is not configured to control the placement of semiconductors onto wafer, such as semiconductors **10** onto semiconductor wafer **20** shown in FIG. **2A**, with a precision appropriate for miniaturized scale such as 1 nm. FIG. **4** illustrates an observed misalignment of semiconductor **10** and semiconductor wafer **20** by a prior art placement tool.

(37) FIG. **4** illustrates when placing individual semiconductor **10** onto wafer **20** illustrated in FIG. **2A**, a shift **404a** and **402b** can take place causing the semiconductor **10** not aligned with the semiconductor wafer **20**. It is observed that such a shift error may be due to a precision of the placement tool **200** is limited and may not satisfy a placement of the semiconductor **10** onto the semiconductor wafer **20**. For example, it is observed that the shift error **404a** and **404b** are between 0.5 to 1 μm , which can cause the conductive regions **152** and **252** are not aligned to affect a performance with the bonded structure **30**.

(38) Improved Placement

(39) For addressing the above-mentioned misalignment when placing individual semiconductors onto a wafer, improvements over the placement tool are made. In some embodiments, the placement tool in accordance with the present disclosure is configured to tilt the pick-up head to form an angle before the semiconductor is placed onto the wafer. In some embodiments, before lowering the semiconductor to the wafer to make a contact, a location of the contact is first detected by the placement tool in accordance with the present disclosure. Based on the detect location, the placement tool in accordance with the present disclosure determines whether the location of the contact is where the semiconductor is supposed to be placed on the wafer such that it is aligned as shown in FIG. **1B**. In those embodiments, if the placement tool determines that a misalignment

exists, it is configured to move accordingly to correct the misalignment. This cycle can continue until the placement tool **200** determines that semiconductor **10** is aligned with the wafer **20**. In some embodiments, the placement tool in accordance with the present disclosure is configured to lay the semiconductor first at the location of the contact and to determine if the contact has been made. In those embodiments, after determining the contact has been made, the placement tool is configured to lay the rest of semiconductor onto the wafer.

(40) FIG. 5 illustrates one example of a placement tool in accordance with the disclosure. As can be seen, the placement tool **500** in this example includes a robotic arm **504**, a pick-up head **506**, a stepper motor **508**, and/or any other components. As shown, the pick-up head **506** is tilt-able such that the semiconductor **10** can be tilted by the placement tool **500** to form an angle **502** with respect to an surface of the wafer **20**. It should be understood although wafer **20** is shown in this example, this is not intended to be limiting. As mentioned above, in various embodiments, the placement tool **500** is used to place semiconductor **10** onto another semiconductor, which may be already placed on the wafer **20**, or may be placed on another semiconductor that is directly or indirectly placed on the wafer **20**.

(41) As can be seen, because the pick-up head **506** is tilted, the semiconductor **10** is also tilted such that a portion of the semiconductor **10** is going to first touch the wafer **20** at a contact point **510a** on the semiconductor **10** before the rest of the semiconductor **10**. As shown, if lowered by the pick-up head **506**, the semiconductor **10** will make a contact with wafer **20** at contact point **510b** on wafer **20**. In implementation, at a predetermined height **520**, the pick-up head **506** is configured to tilt. As shown, the stepper motor **508** of the placement tool **500** in accordance with the present disclosure can be configured to emit one or more optical beams **512** to detect a location of the contact point **510a** and/or **510b**. Because the predetermined height and the angle **502** are known, and because the detected contact location(s) can be known from the optical beam(s), the placement tool **500** is configured to determine the location of the contact **510** when the semiconductor **10** is placed on the wafer **20**. If the determined location of the contact **510** is off from the align structures on the wafer **20** as shown in FIG. 4, the placement tool **500** is configured to correct the pick-up head by the shift error **402a** and/or **402b** shown in FIG. 4. This contact point detection and determination can continue until the placement tool **500** determines that the contact **510** is aligned with corresponding structure on wafer **200**.

(42) FIG. 5B illustrates a top view of the placement of semiconductor shown in FIG. 5A. In implementation, as shown here, one side of the semiconductor is tilted. In those implementation, two contacts can be monitored by placement tool **500**, such as the contact **510** and contact **514**. In those implementation, pick-up head **506** can be adjusted based on the location of contact **510** and the location **514**. For example, the location of contact **510** can be determined by placement tool **500** simultaneously or separately with the determination of the location of the contact **514**. In the case when the location of the contact **510** is determined separately from the location of contact **514**, the pick-up head **506** can be corrected first based on the location of the contact **510**, and can be corrected again based on the location of the contact **514**.

(43) FIG. 5C illustrates that another top view of the placement of semiconductor shown in FIG. 5A. In this view, after the location(s) of one or more contacts is determined and alignment is determined by placement tool **500**, the semiconductor is lowered by the pick-up head **506** onto wafer **20** at contacts **510** and **514** first. This may be referred to a one side positioning contact first. In such a placement, it is understood that an impact is made at the contacts **510** and **514** between semiconductor **10** and wafer **20**. This impact may cause small deformation or cracks on semiconductor **10** and/or wafer **20**. FIGS. 6A-B illustrates such a deformation.

(44) FIG. 6A shows that when the semiconductor **10** is first made contact with the wafer **20** in accordance with the present disclosure, an indentation **602** is caused due to a pressure from a weight of semiconductor **10** and/or a force from a movement of the pick-up head **506**. In view of this indentation, a number of considerations should be factored in when using the placement tool **500** in

accordance with the present disclosure. One consideration is friction. That is when choosing contact **510** and/or **514** for placement, there should be sufficient friction between semiconductor **10** and wafer **20** to prevent sliding. If such a friction is not present at certain parts of the semiconductor **10** and/or wafer **20**, especially on an edge of semiconductor **10**, those parts should be avoided to make the first contact by the pick-up head **506**. Another consideration is deformation in the wafer **20** where the contact(s) are made. This deformation may cause a portion of the wafer **20** is lost (e.g. chipped away) where the contact is made. This deformation should be measured or estimated to ensure the device region on the wafer **20** is not damaged by such a deformation. In one instance, it is observed that the deformation is between 0.1-1 nm at the contact(s), and thereby the contact(s) are made at wafer **20** where it has at least 1 nm depth before the device region is damaged by the contact(s).

(45) FIG. **6B** shows that an indentation can be caused to the semiconductor **10** at the contact(s). It is observed that a void **604** may be formed at the contact(s) such that semiconductor **10** and the wafer **20** is not bonded at void **604**. In implementation, the contact(s) may be selected to account for such a void. For example, if the bonded structure's performance would suffer due to such a void, contact(s) should be avoided where the void is formed. In one instance, it is observed that the void is between 0.2-2 nm in depth.

(46) Referring back to FIG. **5D**, where still another top view of placement of the semiconductor **10** shown in FIG. **5A**. In this view, as can be seen, the rest of semiconductor **10** is laid flat down onto wafer **20** after the contacts **510** and **514** are made.

(47) FIGS. **7A-B** illustrate embodiments for detecting contact locations using the placement tool in accordance with the present disclosure. In FIG. **7A**, optical beams, such as beams **704** and **706**, may be emitted by a stepper motor of a placement tool in accordance with the present disclosure. The beams **704** and **706** may be emitted towards an insulation structure **702** (such as a guard ring). Because the insulation structure **702** has a lower reflection/refraction index than the dielectric layer of semiconductor **10** as shown. Because the beams **704** and **706** are bounced off insulation structure, they hit the wafer **20** at locations shown and then bounce off to be intercepted by the stepper motor again. Because a location of the insulation structure **702** in the semiconductor **10** is known, and because the angles at which the beams are emitted towards semiconductor **10** are known and the angles at which they are intercepted by the bounce off from the wafer **20** are known, the contact locations on the wafer **20** can thus be determined.

(48) FIG. **7B** illustrates that optical beam **708** may be emitted towards an alignment pattern **710** in the semiconductor **10** by the stepper motor. In this design, the optical beam **708** penetrates the semiconductor **10** and diffracts through semiconductor **10**, and then bounce off corresponding alignment pattern **710** on the wafer. In this way, an interference position can be calculated by the stepper motor and an offset can be calculated to determine the contact location.

(49) FIG. **8** illustrates an example method **800** for forming a semiconductor package having a bonded structure in accordance with the disclosure. The method **800** starts at step **802**. At step **802**, a first die is picked up from a substrate using a placement tool. The placement tool comprises a pick-up head and a stepper motor, the pick-up head is configured to be tilt-able and the stepper motor is configured to emit one or more optical beams towards the first die. At step **804**, the first die is moved by the placement tool to a first location above a substrate. At step **806**, the first die is tilted by the pick-up head to form an angle between the first die and the substrate. At step **808**, a contact location of the semiconductor onto the substrate is determined using the stepper motor. At step **810**, a shift is determined based on the contact location. At step **812**, the pick-up head is adjusted, using the stepper motor, to correct the shift-off. At step **814**, it is determined, using stepper motor, that the corrected contact location is aligned with the alignment position on the substrate for the first die. At step **816**, the first die is lower by the pick-up head to make a contact with the substrate at the contact location such that a least one portion of the first die does not contact the substrate. At step **818**, the at least one portion of the first die is laid, using the pick-up

head, onto the substrate.

(50) In accordance with some embodiments of the disclosure, a method is provided. The method includes the following steps: picking up a first die from a substrate using a placement tool, wherein the placement tool comprises a pick-up head and a stepper motor, the pick-up head is configured to be tilt-able and the stepper motor is configured to emit one or more optical beams towards the first die; moving, using the placement tool, the first die to a first location above a substrate; lowering, using the pick-up head, the first die to a predetermined height with respect to the substrate; tilting, using the pick-up head, the first die to form an angle between the first die and the substrate; emitting, using the stepper motor, at least one optical beam towards the first die; determining, using the stepper motor, a contact location of the semiconductor onto the substrate; determining, using the stepper motor, the contact location is shift off from an alignment position on the substrate for the first die; adjusting, using the stepper motor, the pick-up head to correct the shift-off; determining, using stepper motor, the corrected contact location is aligned with the alignment position on the substrate for the first die; lowering, using the pick-up head, the first die to make a contact with the substrate at the contact location such that a least one portion of the first die does not contact the substrate; and laying, using the pick-up head, the at least one portion of the first die onto the substrate.

(51) In accordance with some embodiments of the disclosure, a method for placing a semiconductor onto a substrate is provided. The method includes the following steps: transferring, using a placement tool, the semiconductor along a path over onto the substrate; lowering, using the placement tool, the semiconductor to a predetermined height above the substrate; titling, using the placement tool, the semiconductor, to a predetermined angle; determining, using the placement tool, a first contact point of the semiconductor to the substrate at the predetermined angle; determining, using the placement tool, the first contact point is shift-off from an alignment position on the semiconductor with respect to the substrate; adjusting, using the placement tool, the first contact point to correct the shift-off; and lowering, using the placement tool, the semiconductor to make a first contact with the substrate at the corrected first contact point.

(52) The foregoing merely outlines features of embodiments of the disclosure. Various modifications and alternatives to the described embodiments will be apparent to those skilled in the art in view of the teachings herein. Those skilled in the art will appreciate that equivalent constructions do not depart from the scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method comprising: picking up a first die from a substrate using a placement tool, wherein the placement tool comprises a pick-up head and a stepper motor, the pick-up head is configured to be tilt-able and the stepper motor is configured to emit one or more optical beams towards the first die; moving, using the placement tool, the first die to a first location above a substrate; lowering, using the pick-up head, the first die to a first height with respect to the substrate; tilting, using the pick-up head, the first die to form an angle between the first die and the substrate; emitting, using the stepper motor, at least one optical beam towards the first die; determining, using the stepper motor, a contact location of a semiconductor onto the substrate; determining, using the stepper motor, the contact location is displaced, by a shift-off, from an alignment position on the substrate for the first die; adjusting, using the stepper motor, the pick-up head to correct the shift-off, determining, using stepper motor, the corrected contact location is aligned with the alignment position on the substrate for the first die; lowering, using the pick-up head, the first die to make a contact with the substrate at the contact location such that at least one portion of the first die does not contact the substrate; and laying, using the pick-up head, the at least one portion of the first die

onto the substrate.

2. The method of claim 1, wherein the contact location is a first contact location, and wherein the method further comprises: before the at least one portion of the first die is laid onto the substrate, determining, using the stepper motor, a second contact location of the first die onto the substrate; determining whether the second contact location is shift off from an alignment position on the substrate for the first die corresponding to the second contact location; and adjusting, using the stepper motor, the pick-up head to correct the shift-off of the second contact location.

3. The method of claim 1, wherein the contact location is a first contact location, and wherein the method further comprises: before the at least one portion of the first die is laid onto the substrate, determining, using the stepper motor, a second contact location of the semiconductor onto the substrate; and determining whether the second contact location is shift off from an alignment position on the substrate for the first die corresponding to the second contact location; and, wherein the adjustment of the pick-up head by the stepper motor is further based on the shift-off of the second contact location.

4. The method of claim 1, wherein an impact of the contact at the contact location causes a deformation of the substrate, and the substrate has a first depth greater than the deformation of the substrate.

5. The method of claim 1, wherein the stepper motor is configured to emit two optical beams towards an insulation structure in the first die at two different angles for determining the contact location.

6. The method of claim 1, wherein the stepper motor is configured to emit an optical beam towards an alignment pattern within the semiconductor.

7. The method of claim 1, wherein the substrate is a wafer.

8. The method of claim 1, wherein the substrate is part of a second semiconductor placed directly or indirectly onto a wafer.

9. The method of claim 1, wherein the placement tool is configured to: emit, by the stepper motor, a first optical beam; emit, by the stepper motor, a second optical beam; obtain a difference between a first optical path of the first optical beam and a second optical path of the second optical beam; and determine the contact location based on the difference.

10. The method of claim 1, wherein the placement tool is configured to: emit, by the stepper motor, an optical beam towards a first alignment pattern in the semiconductor, wherein the optical beam is bounced back by a second alignment pattern in the substrate; calculate an interference position based on interference of the optical beam caused by the first alignment pattern and the second alignment pattern; and determine the contact location based on the interference position.

11. A method for placing a semiconductor onto a substrate, transferring, using a placement tool, the semiconductor along a path over onto the substrate; lowering, using the placement tool, the semiconductor to a first height above the substrate; titling, using the placement tool, the semiconductor, to a first angle; determining, using the placement tool, a first contact point of the semiconductor to the substrate at the first angle, wherein the placement tool is configured to emit two optical beams towards an insulation structure in the semiconductor at two different angles for determining the contact point; determining, using the placement tool, the first contact point is shift-off from a first alignment position on the semiconductor with respect to the substrate; adjusting, using the placement tool, the first contact point to correct the shift-off; and lowering, using the placement tool, the semiconductor to make a first contact with the substrate at the corrected first contact point.

12. The method of claim 11, wherein the contact point is a first contact point, and wherein the method further comprises: before at least one portion of the semiconductor is laid onto the substrate, determining, using placement tool, a second contact point of the semiconductor onto the substrate; determining whether the second contact point is shift off from a second alignment position on the semiconductor; and adjusting, using the placement tool, the second contact point by

correcting the shift-off of the second contact point.

13. The method of claim 11, wherein contact point is a first contact point, and wherein the method further comprises: before at least one portion of the semiconductor is laid onto the substrate, determining, placement tool, a second contact point of the semiconductor onto the substrate; and determining whether the second contact point is shift off from a second alignment position on the semiconductor; and, wherein the adjustment of the first contact point is further based on the shift-off of the second contact point.

14. The method of claim 11, wherein an impact of the contact at the contact location causes a deformation of the substrate, and the substrate has a first depth greater than the deformation of the substrate.

15. The method of claim 11, wherein the substrate is a wafer.

16. The method of claim 11, wherein the substrate is part of a second semiconductor placed directly or indirectly onto a wafer.

17. The method of claim 11, wherein the placement tool is configured to: emit, by a stepper motor of the placement tool, a first optical beam; emit, by the stepper motor, a second optical beam; obtaining a difference between a first optical path of the first optical beam and a second optical path of the second optical beam; and determining the contact location based on the difference.

18. The method of claim 11, wherein the placement tool is configured to: emit, by a stepper motor of the placement tool, an optical beam towards a first alignment pattern in the semiconductor, wherein the optical beam is bounced back by a second alignment pattern in the substrate; calculate an interference position based on interference of the optical beam caused by the first alignment pattern and the second alignment pattern; and determine the contact location based on the interference position.

19. The method of claim 14, wherein the first depth is greater than 1 nm.

20. A method for placing a semiconductor onto a substrate, transferring, using a placement tool, the semiconductor along a path over onto the substrate; lowering, using the placement tool, the semiconductor to a first height above the substrate; titling, using the placement tool, the semiconductor, to a first angle; determining, using the placement tool, a first contact point of the semiconductor to the substrate at the first angle, wherein the placement tool is configured to emit an optical beam towards an alignment pattern within the semiconductor; determining, using the placement tool, the first contact point is shift-off from a first alignment position on the semiconductor with respect to the substrate; adjusting, using the placement tool, the first contact point to correct the shift-off; and lowering, using the placement tool, the semiconductor to make a first contact with the substrate at the corrected first contact point.
